POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2017
Programme: BE | Full Marks: 100
Course: Microprocessors | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Compare and contrast Microprocessor and Microcontroller. 7
   b) Draw the block diagram for the internal architecture of Intel 8085 and explain in brief. 8
2. a) Explain in brief the different addressing modes of 8085 with proper examples. 7
   b) Write an 8085 ALP that checks an array of ten memory location starting from C000H for odd numbers including zero and transfers the odd numbers in memory location starting from C0EH. 8
3. a) Explain bus structure of a Microprocessor. What do you mean by Synchronous and Asynchronous Bus? 7
   b) With a suitable diagram explain the read and write operation with a SRAM cell. 8
4. a) Draw the block diagram of 8255 PPI and explain in brief. 7
   b) What are the significance of assembler's directive in microprocessor? List all the assembler's directive of 8086 microprocessor and explain them with example. 8
5. a) How interrupt processing occurs in a microprocessor? Explain vector chain and polled interrupt. 7
   b) Write an assembly language program in 8086 which reads a string from a keyboard and then displays the string in the reverse order. 8
6. a) Write a program in 8086 to find whether a number is positive or negative. 7
   b) RS232 forms a basis for serial data communication standard. Explain with diagram. 8
7. Write short notes on: (Any two) 2x5
   a) Memory mapped I/O versus I/O mapped I/O
   b) Flags in 8085
   c) Serial Communication

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